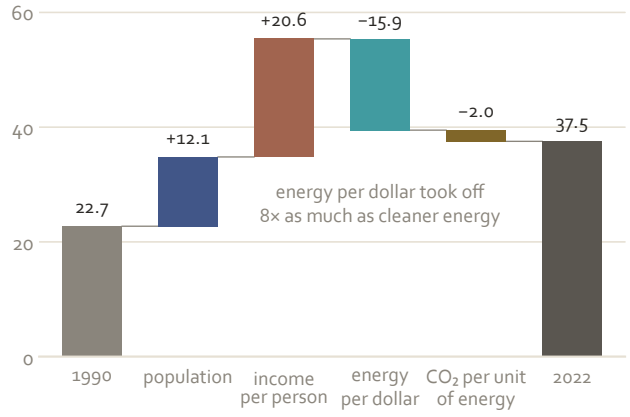
Since 1990 rising incomes added 20.6 Gt to the world's yearly CO_2; using less energy per dollar took off 15.9 Gt, a cleaner energy mix only 2.0 Gt

Fossil and industry CO_2 rose from 22.7 to 37.5 Gt a year, 1990–2022, split exactly by the Kaya identity (LMDI)

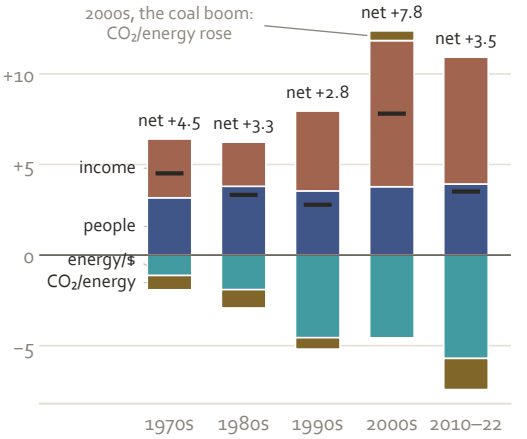
a The rise of 14.8 Gt from 1990 to 2022, by factor

gigatonnes of CO_2 a year



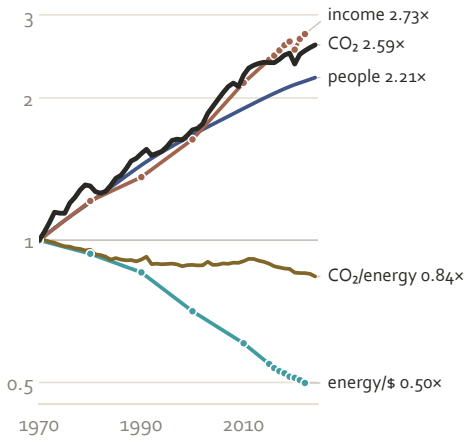
b Each decade: growth against efficiency

change in yearly CO_2, Gt per decade



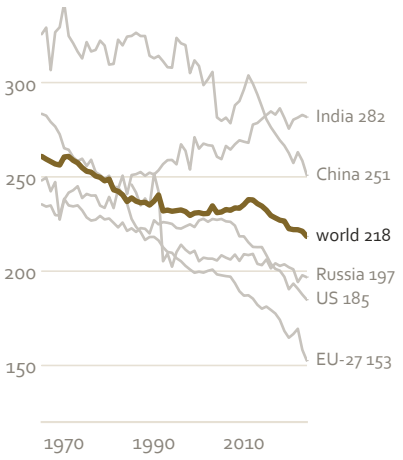
c The four factors

index, 1970 = 1 (log) · dots: GDP years



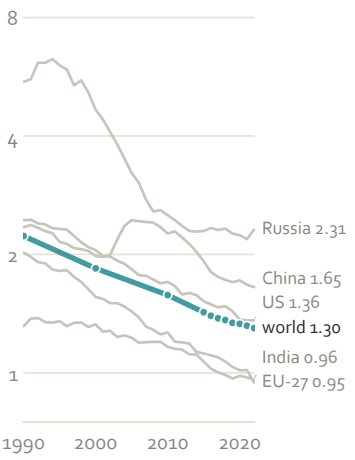
d CO_2 per unit of energy

grams of CO_2 per kWh of primary energy



e Energy per dollar

kWh of primary energy per dollar of GDP (log)



Global Carbon Budget (2025), Energy Institute, Maddison Project (2023), UN WPP; OWID CO_2 dataset, CC BY 4.0, downloaded 2026-10-12; Kaya (1990), LMDI after Ang (2004) · sus_kaya.py · git 4c16514